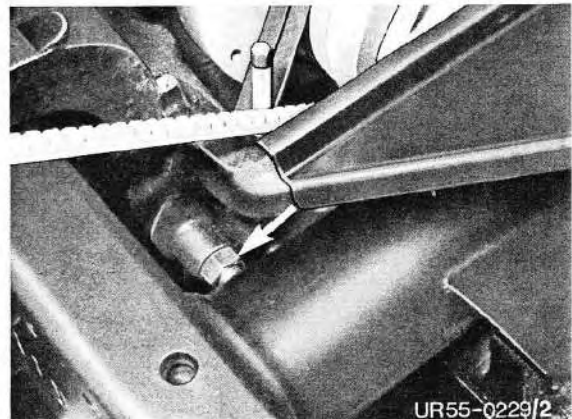
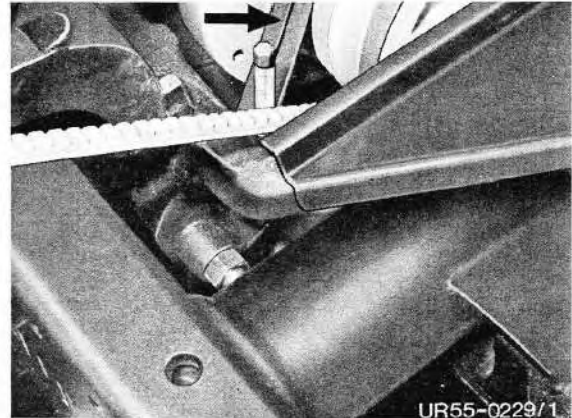


V-belt, hydraulic pump

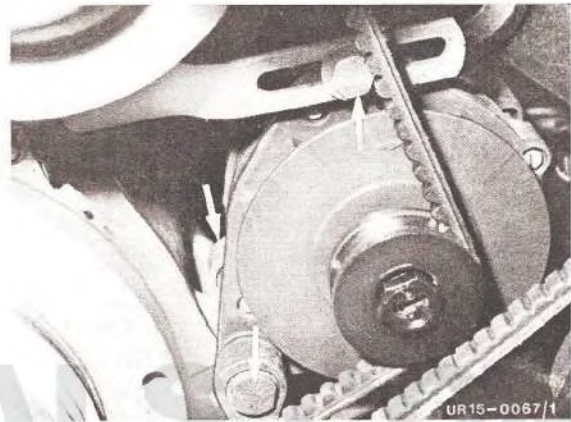
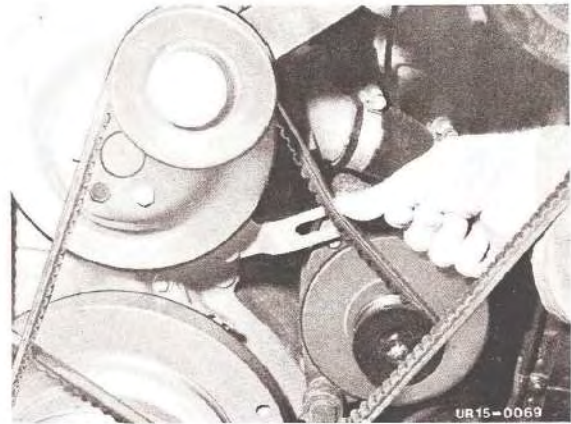
- 1 Prop up engine hood and remove front panel.
- 2 Check V-belt for condition and tension, tightening or exchanging if necessary.
See 1.1/1 for amount of yield.
- 3 Remove V-belts of intermediate pump drive, air compressor and power steering pump, referring to 2.1/1 to 2.4/1.
- 4 Release bolt holding pump bracket.
- 5 Release tensioning screw and remove V-belt.
- 6 Install in reverse sequence.

Note: V-belts must be exchanged in sets.



V-belt, coolant pump/generator

- 1 Prop up engine hood and remove front panel.
- 2 Check V-belt for condition and tension, tightening or exchanging if necessary.
See 1.1/1 for amount of yield.
- 3 Remove V-belts of intermediate fan drive, air compressor, power steering pump and hydraulic pump, referring to 2.1/1 and 2.5/1.
- 4 Release tension bracket and swivel bearing.
- 5 Remove V-belt.
- 6 Install in reverse sequence.



Contents

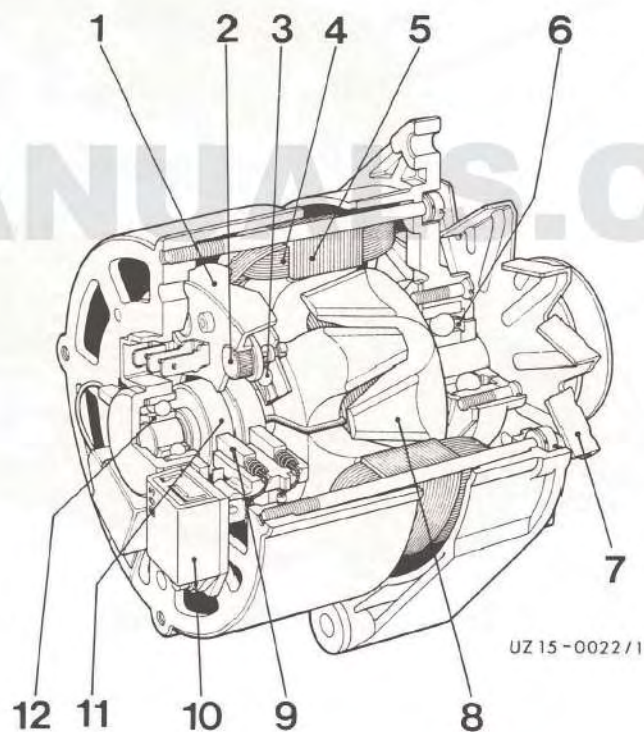
Chapter	Page
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3 Reconditioning starter	
Removal and installation of starter	3.1/1
Exchange of carbon brushes	3.2/1

Technical Data

Electrical System	12 V standard ¹⁾	12 V/SA 35 496 ¹⁾ Standard ²⁾	24 Volt SA 35 496	24 Volt SA 35 496
Manufacturer	Bosch			
Design	Three-phase current			
Power output W	490	770	756	1540
Rated voltage V	14		28	
Current A	35	55	27	55
Maximum speed 1/min	9000		11000	
Begins to charge at 1/min	1000		1180	
Transmission ratio engine/alternator	1: 1,85			1: 3,268
Marking	K 1 - 14 V 16/35 A	K 1 - 14 V 23/55 A	K 1 - 28 V 7/27 A	N 1 - 28 V 10/55 A

1) To engine end No. 077 043.

2) From engine end No. 077 044 installed as standard, can be retrofitted.

Sectional View**Three-phase Alternator**

- | | |
|-----------------------|-----------------------------|
| 1 Heat sink | 7 Fan |
| 2 Power diode | 8 Interlocking-finger rotor |
| 3 Exciter diode | 9 Carbon brush |
| 4 Stator winding | 10 Regulator |
| 5 Stator lamination | 11 Slip ring |
| 6 Drive bearing mount | 12 Slip ring bearing |

15.8 General

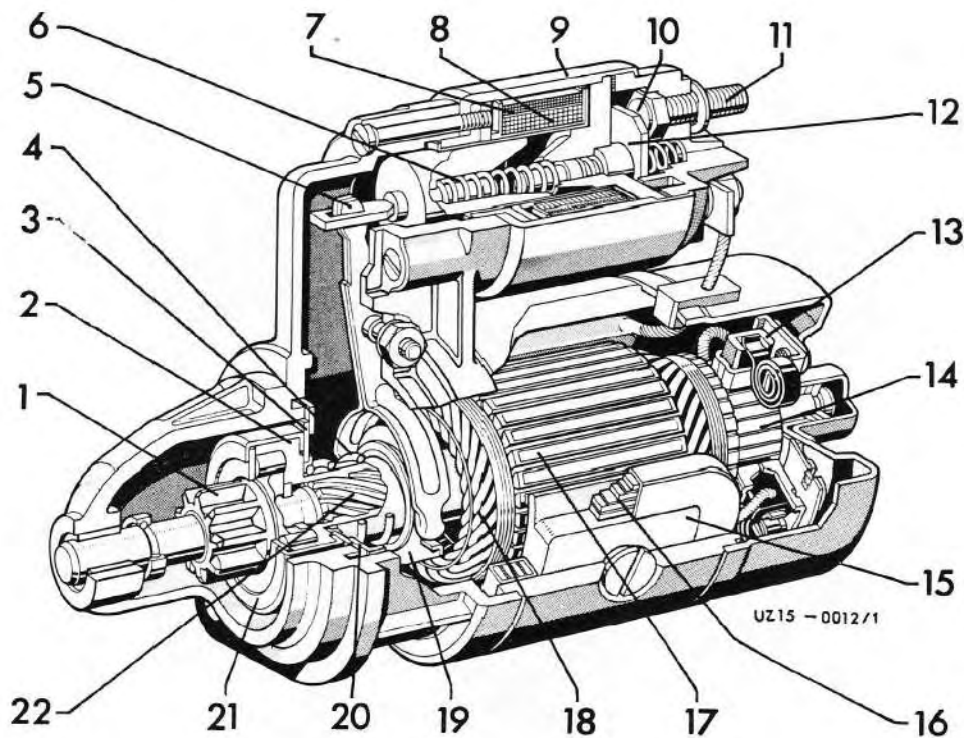
352/352 A

Technical Data

Installation	12 Volt (standard)	24 Volt (SA 82 983)	
Type	IF → 12 V 3 kW	IF → 24 V 4 kW	
Manufacturer	Bosch		
Design	Screw-push drive		
Power output	kW	3	4
Rated voltage	V	12	24
Direction of rotation	clockwise		
Number of teeth on pinion	9		

Note: The previous ID starter has been replaced from chassis end No. 073 856 or engine end No. 588 644 by the IF starter.

Sectional View



Starter

- | | | | |
|----|-----------------|----|------------------------------|
| 1 | Pinion | 12 | Contact plate |
| 2 | Driver | 13 | Carbon brush |
| 3 | Brake disk | 14 | Commutator |
| 4 | Meshing spring | 15 | Pole shoe |
| 5 | Shift lever | 16 | Exciter winding |
| 6 | Return spring | 17 | Starter armature |
| 7 | Hold-in winding | 18 | Armature winding |
| 8 | Draw-in winding | 19 | Guide ring |
| 9 | Solenoid relay | 20 | Stop |
| 10 | Relay contact | 21 | Over-running clutch |
| 11 | Terminal | 22 | Coarse-thread armature shaft |

Tightening torques

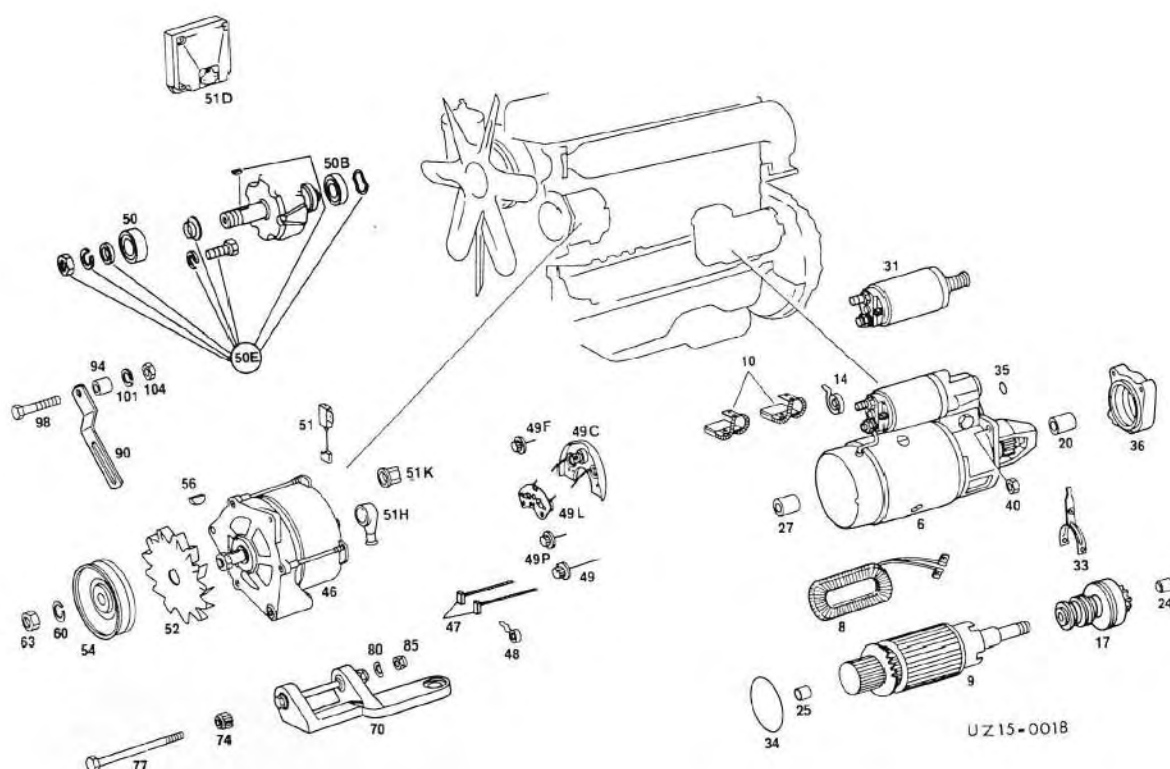
Location	Thread material	Nm
Starter to clutch casing	M 10	55 + 10
Stud for starter to clutch casing	M 10	10 to 20
Generator clamping plate to crankcase	M 10	40 ± 5
Fastening bolt, generator to bracket	M 12	40 ± 5
Clamping bolt, generator to crankcase	M 10	40 ± 5
Pulley to generator	M 14	35 + 10

Consumibles

Cons. No.	Description	Part No.
1	Sealant Loctite No. 573	001 989 45 20
2	Sealant Curil K2	regular

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Exploded view



Starter and generator

6	Starter	50	Grooved ball bearing
8	Exciter coil	50B	Ball bearing (slip ring side)
9	Armature	50E	Repair kit
10	Carbon brush set	51	Condenser
14	Spring	51D	Regulator and but-out
17	Starter pinion	51H	Boot
20	Bushing in drive bearing	51K	Boot
24	Busher (Starter pinion)	52	Fan
25	Bushing (in intermediate bearing)	54	Pulley
27	Bushing (in commutator bearing)	56	Woodruff key
31	Solenoid switch	60	Lock washer
33	Control lever	63	Nut
34	Seal ring	70	Bracket
35	Seal ring	74	Bushing
36	Intermediate piece	77	Screw
40	Nut	80	Spring washer
46	Generator	85	Nut
49	Diode neg.	90	Bracket
49C	Rectifier	94	Spacer tube
49F	Diode pos.	98	Screw
49L	Diode plate	101	Washer
49P	Diode exciter	104	Nut

Removal

- 1 Remove engine bonnet outwards and from cover forwards.
- 2 Disconnect battery.

Note: Disconnect (–) terminal first, then (+). Switch off the battery master switch if fitted.

- 3 Release bolts in tensioner and jockey pulley, remove Vee belt.
- 4 Mark electrical connections on generator and remove, withdraw bolts from jockey pulley and tensioner and remove generator with distance piece.

- 5 Remove fan with belt pulley.
- 6 Clean, test and if necessary replace all parts.

Installation

- 1 Fit fan with belt pulley and tighten. Tightening torque, see 1.3/1.
- 2 Install generator with distance piece and insert bolts in jockey pulley and tensioner.

- 3 Fit electrical connections to generator according to markings.

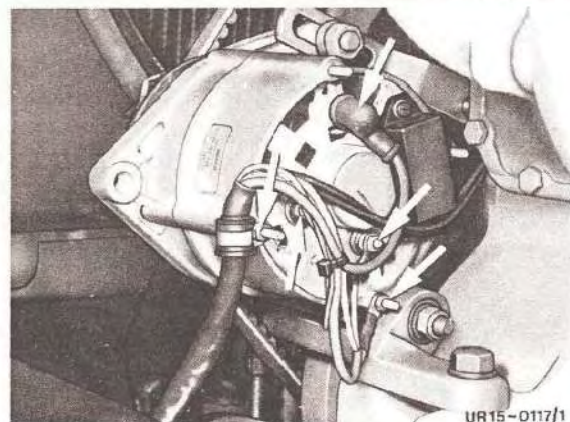
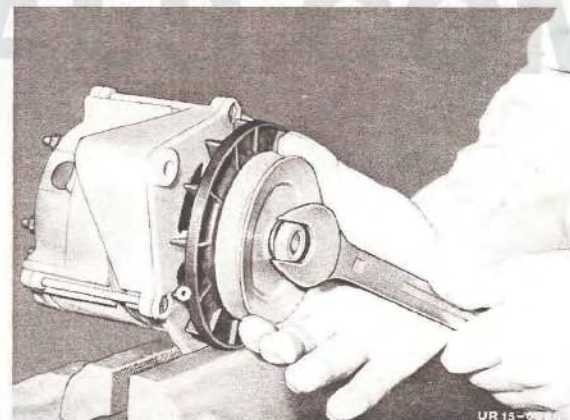
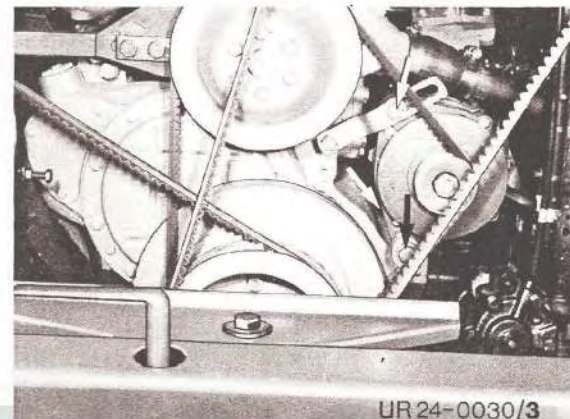
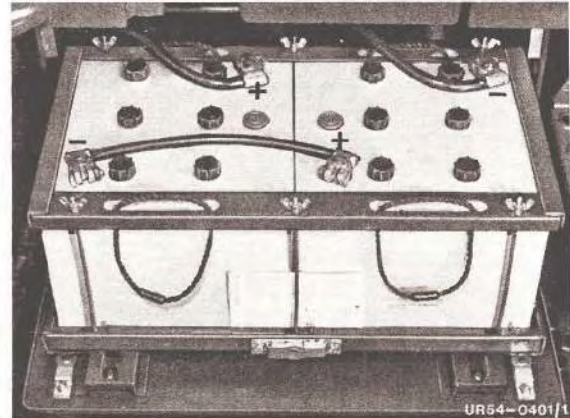
Note: Except for generator N 1-28 V 55 A 25 from engine end No. ... 538 581 the regulator is built into the generator. On fitting a new generator the cable "D" (colour code black) and "DF" (colour code brown) are to be cut out of the generator and the multi-pin plug from the regulator. The regulator can be dismantled.

- 4 Fit Vee belt and tension, see 13.14-2.6/1.

- 5 Connect battery.

Note: Connect (+) terminal first, then (–). Switch on the battery master switch if fitted.

- 6 Install front cover and engine hood.



Removal and installation

- 1 Detach engine hood outside and in driver's cab.
- 2 Disconnect battery.

Note: Disconnect first (–), then (+) terminal. If a main battery switch is fitted, turn off first.

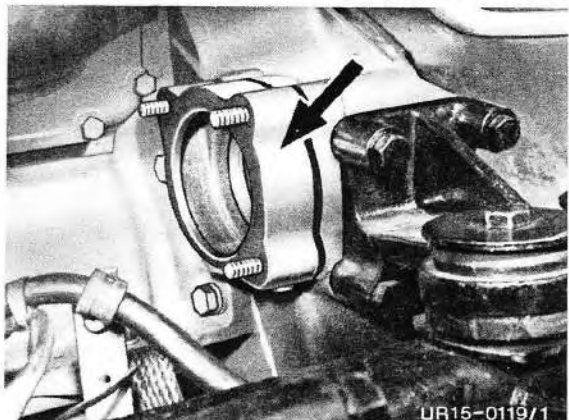
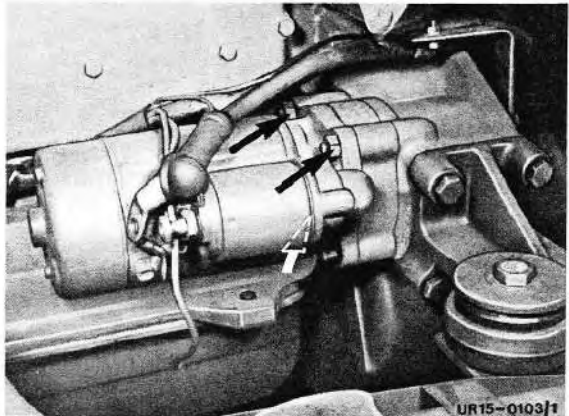
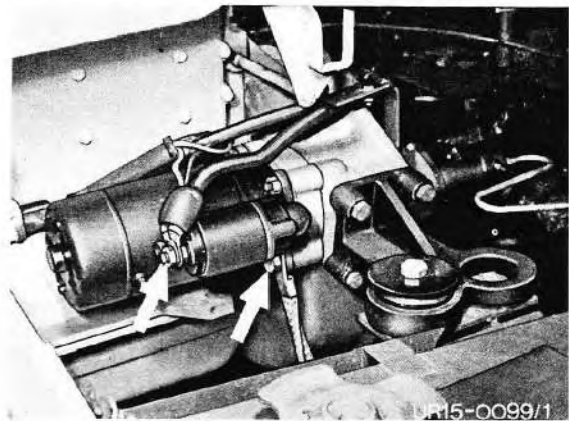
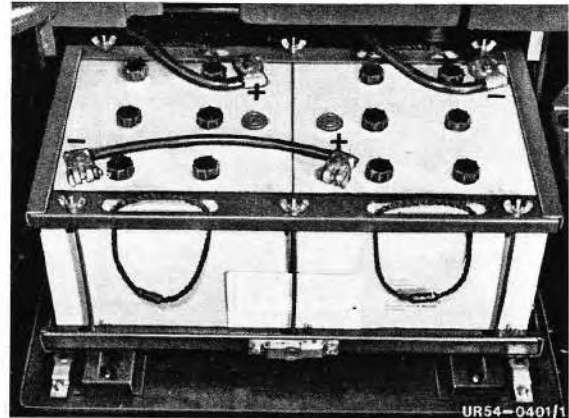
- 3 Disconnect wiring from starter.

- 4 Detach starter from clutch housing.
- 5 Check starter ring gear and exhcnage, if necessary, referring to 03.8-4.1/1.

- 6 Take intermediate piece and make sure that studs are firmly seated in clutch housing. Refer to tightening torques, 1.3/1.

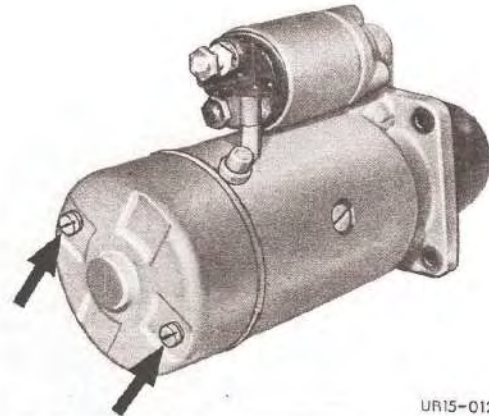
- 7 Exchange carbon brushes, referring to 3.2/1.
- 8 Clean, check and exchange if necessary all parts.
- 9 Installation in opposite sequence to removal.

Note: Fit sealant No. 1 to flange surfaces and sealant No. 2 to nuts and tighten. Tightening torque, see 1.3/1.



Replacement

- 1 Remove starter, referring to 3.1/1.
- 2 Unscrew and remove cover.



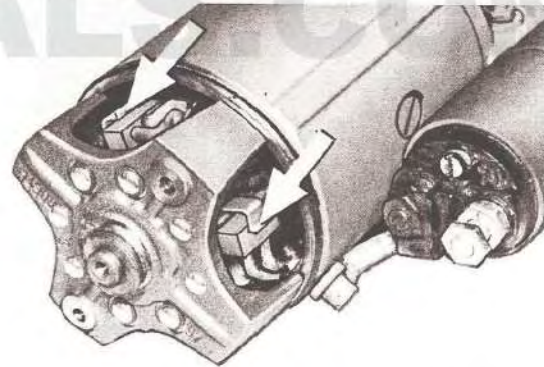
UR15-0120/1

- 3 Check carbon brushes for wear and condition.

Note: Carbon brushes must not be oily, nor must wiring touch holder.

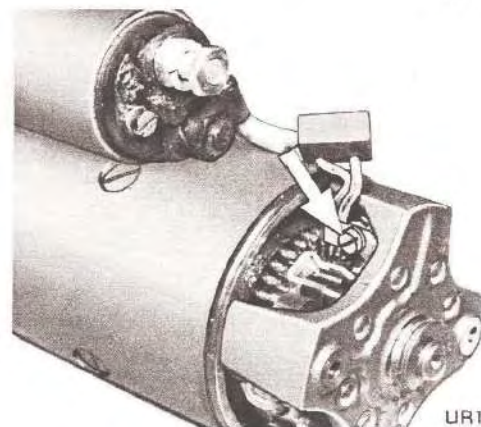


- 4 If brushes need replacing, pull back spring clip and withdraw brush from holder.



UR15-0121/1

- 5 Unscrew brushes. Clean commutator and surrounds.
- 6 Fasten new brushes.
- 7 Insert carbon brushes into holder, making sure that they do not stick.
- 8 Fasten cover with new dealing ring.
- 9 Install starter, referring to 3.1/1.



UR15-0122/1

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5 Venting engine	5.1/1

Oil pump test

Delivery, measured with SAE 10 oil, at 50°C oil temperature, 4.0 bar back pressure	300/min	l/min	6.0
	1400/min	l/min	43.0
Opening pressure of relief valve		bar	5.2 ± 0.5
Oil temperature		°C	max. 120
Oil pressure gage reading for new engine (operating temperature)	idling at about 700/min	bar	1.2
	at max. speed	bar	5
Oil pressure gage reading for used engine (operating temperature)	idling at about 700/min	bar	0.6
	at max. speed	bar	2.5

Springs for oil pressure relief valves

	OD mm	Wire gage mm	Setting bar	Length extended mm	Spring preloaded Length Load mm N		Spring at final load Length Load mm N	
In oil pump	9.3	1.7	5.2 ± 0.5	49.4	45.4	43.9	36.2	145
In oil filter body	16.5	1.5	1.96 ^{+0.6} _{-0.2}	66	31	45 ± 3	21 ± 0.5	57
In oil cooler	17.0	1.25	1.47-1.86	25.6	12	20	9.0	24.4

Oil pump

Bore in oil pump body	mm	<u>17.018</u> 17.000
Drive shaft dia.	mm	<u>16.984</u> 16.976
Oil pump shaft dia.	mm	<u>15.039</u> 15.028
Height of gear wheel	mm	<u>39.975</u> 39.950
Width of gear wheel	mm	<u>39.975</u> 39.946
Backlash	mm	0.15 to 0.25
Minimum axial spacing between drive gear and end face of oil pump body	mm	0.04
Radial play of drive shaft	mm	0.016 to 0.042
Radial play between oil pump gear and oil pump shaft	mm	0.011 to 0.040
Overlap of oil pump shaft in body	mm	0.010 to 0.039
Radial play of oil pump gears between body and gear wheel	mm	0.030 to 0.105
End play of oil pump gears between cover and gear wheel	mm	0.025 to 0.089
Backlash of oil pump gears	mm	0.15 to 0.25
Backlash of oil pump drive gears (helical gears)	mm	0.096 to 0.128

Special tools

Cons. No.	Designation	Special tool No.	Tool set
1	Oil reservoir	352 589 11 63 00	B

Necessary materials

Cons. No.	Designation	Part No.
1	Adhesive: National adhesive 5821	commercially available
2	Locking compound: Loctite Nr. 270	002 989 93 71
3	Sealing compound: Curil T	001 989 37 20
4	Sealing compound: Curil K 2	commercially available
5	Sealing compound: Dirko-Transparent	001 989 29 20

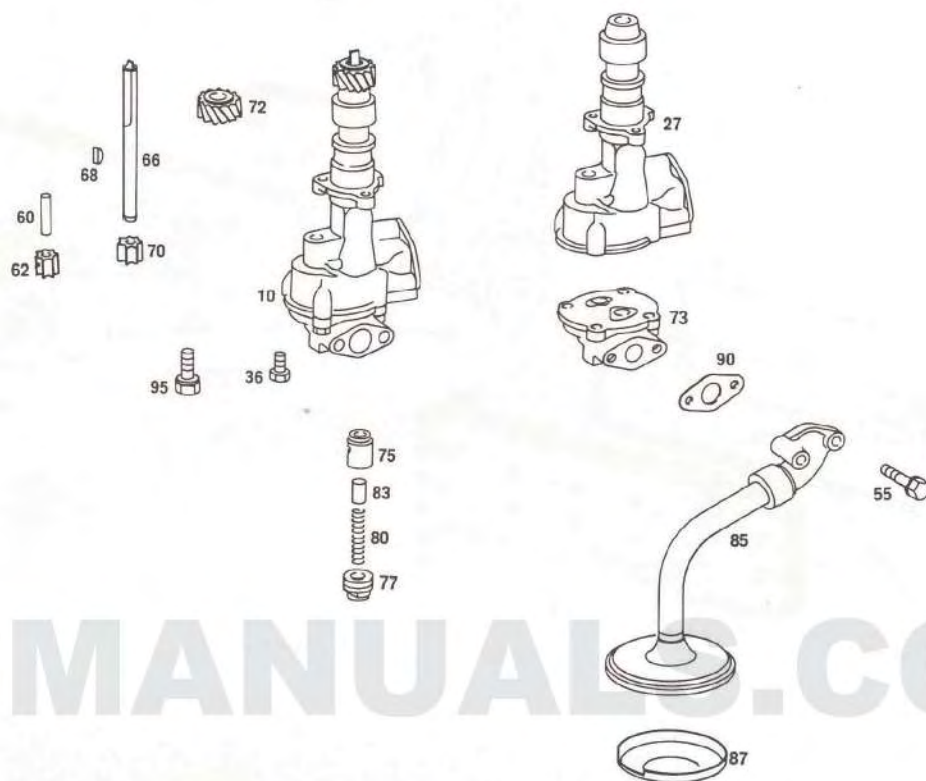
Tightening Torques

Location	Thread	Nm
Oil pan	M 6	8
	M 8	9
Oil drain plug to oil pan	M 26	50 + 10
Oil cooler to crankcase	M 8	35
Pressure relief valve to oil cooler	M 22	60 + 10
Filter bowl to housing cover ¹⁾	M 10	40 to 45
Oil pump to crankcase	M 8	35
Cover to oil pump		35
Oil pressure relief valve to oil pump ²⁾	M 24	15 + 10
Suction strainer to oil pump cover		20 to 25
Oil filter support to crankcase	M 10	60
Oil filter bowl to oil filter support	M 12	40 + 10
Pressure relief valve to oil filter	M 18	40
Plug for oil pressure relief valve – oil filter	M 24	60 + 20
Drain plug oil filter bowl	M 10	15±2
Exhaust manifold to cylinder head	M 10	30
Exhaust manifold to cylinder head 2-hole metal strap	M 10	25
Flanged manifold to exhaust pipe	M 10	45
Starter motor to clutch housing	M 10	55 + 10
Stud for starter to clutch housing	M 10	10 to 20

1) Only engine model 352 A

2) Insert screw plug with locking compound No. 2.

Exploded view



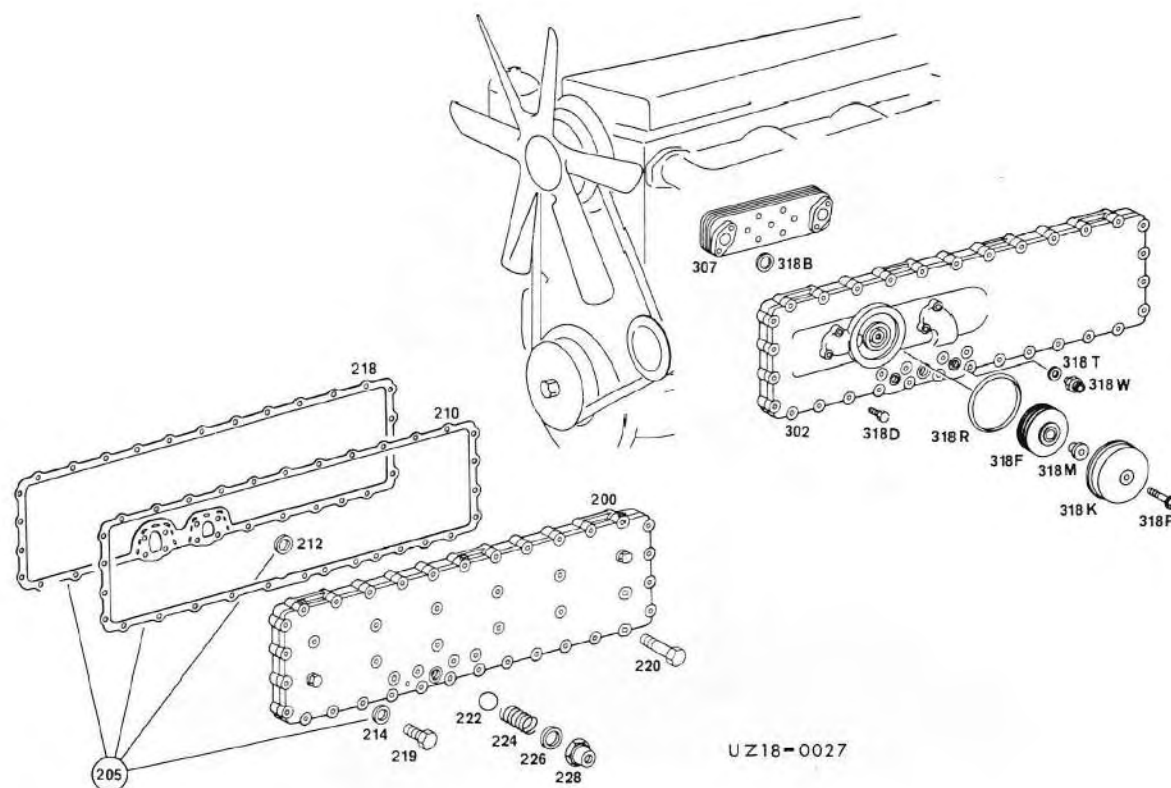
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Oil pump

- 10 Oil pump
- 27 Body
- 36 Bolt
- 55 Bolt
- 60 Oil pump shaft
- 62 Gear wheel
- 66 Drive shaft
- 68 Woodruff key
- 70 Gear wheel
- 72 Helical gear
- 73 Cover
- 75 Body
- 77 Screw plug
- 80 Spring
- 83 Piston
- 85 Suction pipe
- 87 Strainer
- 90 Gasket
- 95 Bolt

352/352 A

Exploded view

Plate-oil cooler
engine 352

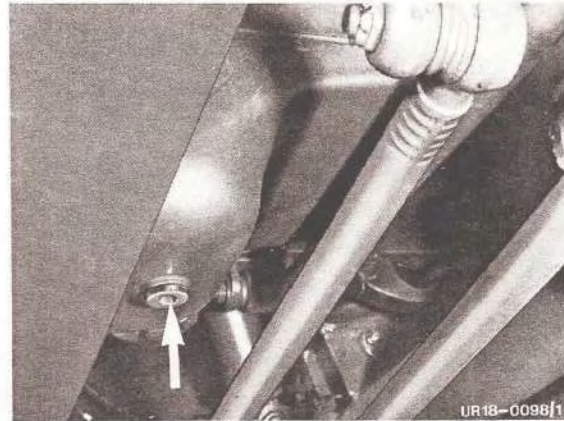
- 300 Oil cooler
- 305 Repair kit
- 310 Gasket
- 312 Seal ring
- 314 Seal ring
- 318 Gasket
- 319 Screw
- 320 Screw
- 322 Ball
- 324 Spring
- 326 Seal ring
- 328 Screw plug

Oil cooler with screen filter
engine 352 A

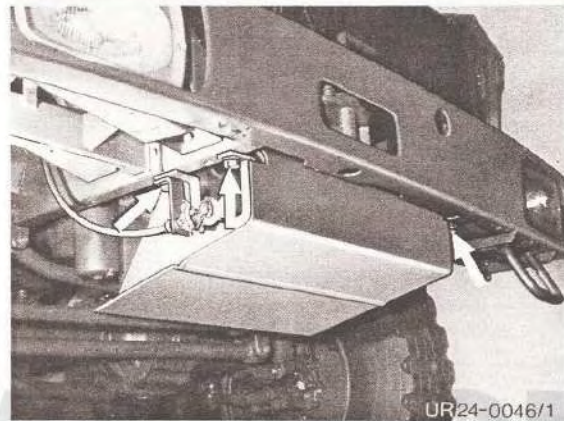
- 302 Cover
- 307 Oil cooler
- 318B Seal ring
- 318D Screw
- 318F Filter element
- 318K Filter cover
- 318M Gasket
- 318P Screw
- 318R Seal ring
- 318T Seal ring
- 318W Screw fitting

Removal

1 Drain off engine oil.



2 Detach underride guard from frame.



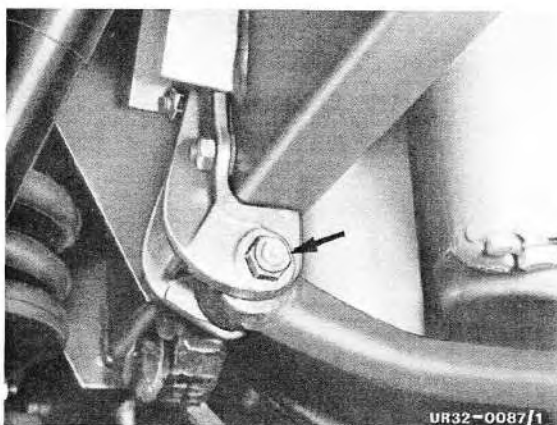
3 Unscrew oil dipstick tube and air compressor oil return line at oil pan.

4 For pressure-charged engine, unscrew turbo-charger oil return line at oil pan.

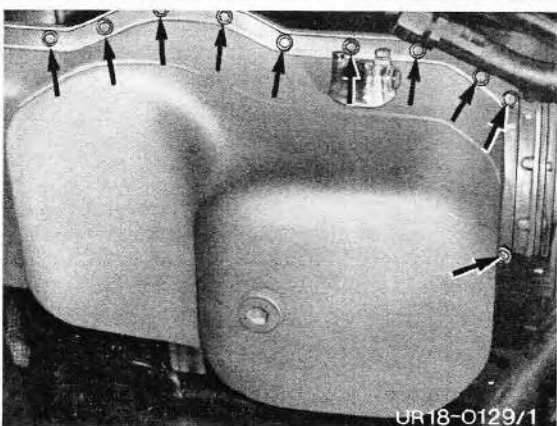


5 Disconnect cable at oil pressure pickup. Detach air compressor and injection pump oil lines at oil filter body. Unscrew oil filter assembly from crankcase.



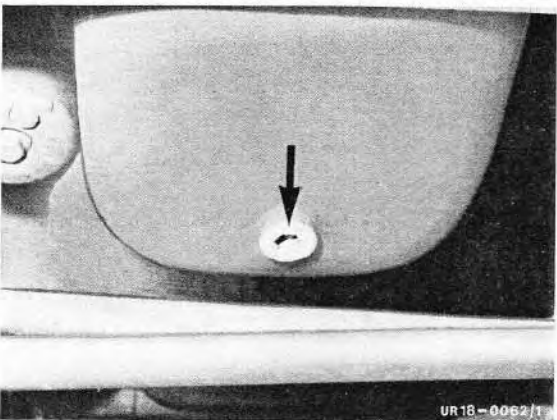


- 6 Remove upper transverse link support.



- 7 Unscrew sump retaining bolts and remove sump.
8 Clean sealing surfaces on sump and crankcase.

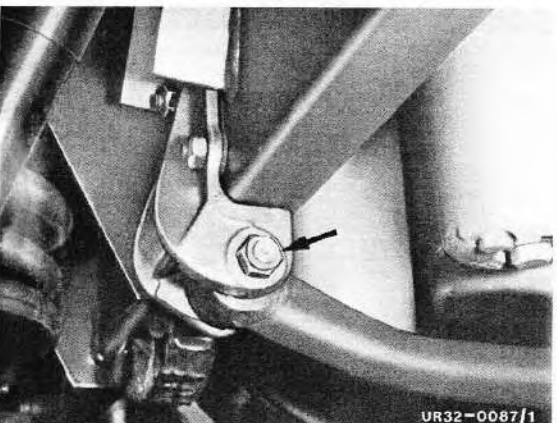
Installation



- 1 Using adhesive No. 1 cement new gasket to the crankcase or bearing cap and timing case sealing surface. In addition the protrusion area of the longitudinal gasket to the preformed gaskets is to be coated with sealant No. 5.

- 2 Carefully press pan on to crankcase and tighten. Tightening torque, see 1.2/1.

- 3 Fit oil drain plug with new washer. Tightening torque, see 1.2/1.



- 4 Refit upper transverse link support.

352/352 A

5 Lightly grease new gasket and attach to mating surface on the crankcase.

Note: Filter housing gasket must lie correctly. If incorrectly fitted, the oil circuit will be broken.

6 Fit oil filter to the crankcase.
Tightening torque, see 1.2/1.

7 Fit oil pipes from compressor and injection pump as well as cable for oil pressure sender.

8 With a turbocharged engine fit oil return pipe for turbocharger to pan.

9 Fit oil return pipe from the compressor and guide tube for oil dipstick to the pan.

10 Fit underrun guard to vehicle frame.

